

Supplementary Information for *Physics-Based Transfer Learning for Neural Network Surrogates of Metal–Insulator–Metal Absorbers*

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This document contains the full ablation tables, baseline comparisons, and supporting analyses referenced in the main text. The supplement comprises Sections S1–S19 (S17, the TMM geometry-constraint sensitivity; S18, the Reviewer #2-requested figures; and S19, the correction change-log, were added in revision).

S1 Random Pre-training Baseline (full table)

To verify that transfer learning benefits arise from physics content rather than pre-training regularization effects, we compared M_{TL} against M_{rand} (pre-trained on uniformly random absorptance spectra). Table S1 reports the full per-seed comparison.

Table S1: Random pre-training baseline (M_{rand}) vs. physics-based models on Structure A (mean $\pm$ std, 10 seeds). M_{rand} (pre-trained on uniformly random absorptance spectra with no physical content) is statistically indistinguishable from training from scratch (M_0), whereas TMM pre-training (M_{TL}) reduces MAE by 20–50%, confirming that the PBTB benefit arises from genuine physics knowledge, not mere weight initialization or regularization.

n	M_0	M_{TL}	M_{rand}
50	8.87 ± 1.01	4.46 ± 0.58	8.79 ± 1.03
100	5.59 ± 0.43	3.26 ± 0.19	5.83 ± 0.36
200	3.63 ± 0.30	2.45 ± 0.07	3.83 ± 0.15
350	2.45 ± 0.14	1.97 ± 0.03	2.50 ± 0.03

The gap between M_{TL} and M_{rand} is 0.5–4.3 pp across training sizes (largest at small n), confirming that the TMM pre-training signal contains genuine physics knowledge that random initialization cannot replicate; M_{rand} itself tracks M_0 to within $\sim 5\%$.

S2 TMM Pre-training Data Size Sensitivity

Overhead is computed as (TMM generation + pre-training time) / RCWA generation time for 500 total samples ($\sim 15,000$ s at ~ 30 s/sample on RTX 4070 Ti SUPER). The benefit increases monotonically with N_{TMM} up to the largest size tested (5,000): the M_{TL} -vs- M_0 improvement at $n = 100$ grows from +26% (500) to +45% (5,000), with $N_{\text{TMM}} = 2,000$ already capturing most of the gain (+38%); even 500 samples provide a +26% improvement at $< 0.5\%$ overhead. Pre-training times are measured with the GPU full-batch implementation.

Table S2: Effect of TMM pre-training data size (Structure A, 3 seeds per condition).

N_{TMM}	Gen. (s)	Pre-train (s)	TMM val MAE	Fine-tuned MAE (%)		Overhead (%)
				$n = 100$	$n = 350$	
Scratch	—	—	—	5.80 ± 0.31	2.35 ± 0.09	—
500	1.9	9	1.70%	4.30 ± 0.16	2.47 ± 0.07	0.1%
1,000	3.8	18	1.59%	4.03 ± 0.26	2.33 ± 0.06	0.1%
2,000	7.8	35	0.77%	3.57 ± 0.16	2.11 ± 0.02	0.3%
5,000	18.9	88	0.45%	3.19 ± 0.06	1.99 ± 0.04	0.7%

S3 Learning Rate Ablation

A potential confound is that M_{TL} uses a lower fine-tuning learning rate (3×10^{-4}) than M_0 (10^{-3}). We conducted a full factorial sweep (2 models $\times$ 2 learning rates $\times$ 4 data sizes $\times$ 3 seeds = 48 runs).

Table S3: Learning-rate ablation across all three structures (3 seeds): M_{TL} -vs- M_0 weight-transfer benefit (%) at matched LR (10^{-3} , which controls the LR confound) and at the deployment fine-tuning LR (3×10^{-4}).

Struct.	LR	$n=50$	$n=100$	$n=200$	$n=350$
A	10^{-3}	+38	+40	+29	+16
	3×10^{-4}	+53	+52	+46	+34
B	10^{-3}	+26	+16	+15	+11
	3×10^{-4}	+38	+23	+16	+12
C	10^{-3}	+8	+2	+4	0
	3×10^{-4}	+16	+9	+5	+4

At the matched learning rate (10^{-3}) the weight-transfer benefit stays clearly positive for Structures A (+16 to +40%) and B (+11 to +26%) and is marginal for the low-fidelity Structure C (marginally negative to +8% at matched LR, dipping to $\approx -0.5\%$ at the largest pool), confirming that the benefit is *not* an artifact of the lower fine-tuning rate used for transfer; the deployment rate (3×10^{-4}) further improves it by better preserving the pre-trained features. The matched-LR benefit is graded by source fidelity ($A > B > C$), consistent with the fidelity-graded pattern of the main results. The Structure-C rows in this ablation use the isotropic single-output Structure-C variant, whose M_0 baseline differs from the main dual-polarization Structure-C, so they are not directly comparable to the +9.7% benefit of the main-text Structure-C table. These ablations use a training batch of 4,096 for efficiency; because M_0 and M_{TL} are trained identically within each cell, the within-cell M_{TL} -vs- M_0 contrast is not biased by the batch-size choice.

S4 Architecture Ablation: ResNet vs. MLP

To rule out that physics features and TL primarily compensate for the ResNet backbone’s capacity, we compare the ResNet-256-4 backbone (566,529 $\approx$ 567 K parameters) against a plain MLP without residual connections or LayerNorm (an input projection to 256 followed by three 256-unit hidden layers and a 128-unit head, SiLU activations; 233,473 $\approx$ 233 K parameters) under identical training protocols.

This ablation uses the isotropic single-output Structure-C variant, whose M_0 baseline differs from the main dual-polarization Structure-C (the main-text Structure-C table), so its Structure-C cells are not directly comparable to the main +9.7% benefit. The ResNet backbone yields

Table S4: Architecture ablation across all three structures (3 seeds): M_{TL} -vs- M_0 benefit (%) with the ResNet-256-4 backbone vs. a plain MLP (three 256-unit hidden layers; Section S4).

Struct.	Arch	$n=50$	$n=100$	$n=200$	$n=350$
A	ResNet	+52	+49	+37	+25
	MLP	+53	+47	+32	+21
B	ResNet	+29	+17	+17	+12
	MLP	+37	+16	-2	-9
C	ResNet	+19	+6	-1	-1
	MLP	+35	+8	-5	-7

a robustly positive transfer benefit for Structures A and B and a marginal-to-neutral outcome for the lowest-fidelity Structure C at the largest training sizes (Structure A +25 to +52%; Structure B +12 to +29%; Structure C -1 to +19%, dipping marginally negative with the ResNet, not only the MLP). With a plain MLP the benefit holds for the high-fidelity Structure A but turns *negative* at large n for Structures B and C (-9% and -7% at $n = 350$), indicating that for moderate- and low-fidelity sources the residual/normalized ResNet architecture is needed for stable transfer. We report this architecture dependence as a robustness caveat: PBTL’s benefit is not architecture-agnostic, and the residual backbone used throughout this work is the stable choice.

S5 Permutation Feature Importance

We quantify the contribution of each physics feature via permutation importance: for each of the 17 features (Structure A), we randomly shuffle its values (10 repetitions) while holding all other features fixed, and measure the resulting MAE increase.

Table S5: Top 10 physics features by permutation importance (M_{phys} , Structure A, $n = 350$).

Rank	Feature	Category	ΔMAE (%)
1	$f_{\text{rect}} = W_x W_y / P^2$	Fill fraction	+8.88
2	$\cos(\varphi_{\text{SiO}_2})$	Cavity resonance	+5.24
3	$\sin(\varphi_{\text{SiO}_2})$	Cavity resonance	+3.90
4	$\sin(\varphi_{\text{TiO}_2})$	Cavity resonance	+3.71
5	t_1/δ	Skin depth ratio	+3.54
6	P/λ	Sub-wavelength	+2.41
7	W_y/W_x	Geometry	+2.08
8	t_{mid}/δ	Skin depth ratio	+1.90
9	α_{metal}	Geometry	+1.51
10	$\cos(\varphi_{\text{TiO}_2})$	Cavity resonance	+1.36

Table S6: Category-level aggregation of permutation feature importance (M_{phys} , Structure A, $n = 350$). Cavity resonance and fill fraction together account for $\sim 60\%$ of total importance.

Category	Features	M_{phys}	Total	% of Total
Cavity Resonance	4	+14.21%		37%
Fill Fraction	2	+9.03%		23%
Skin Depth Ratio	3	+5.78%		15%
Angle & Geometry	3	+4.62%		12%
Sub-wavelength Ratio	3	+4.11%		11%
Optical Path	2	+0.72%		< 2%

Feature importance rankings are nearly identical between M_{phys} and $M_{\text{TL+phys}}$, confirming that feature utility is independent of whether transfer learning was applied.

S6 TMM-as-Input Baseline

An alternative to encoding TMM knowledge through weight initialization (PBTL) is to supply the TMM prediction directly as an additional input feature: $M_{\text{TMM-in}}$ appends the scalar $A_{\text{TMM}}(\mathbf{x}, \lambda)$ to the geometry input and trains from scratch.

Table S7: TMM-as-input baseline vs. PBTL variants (Structure A, 5 seeds).

n	M_0	$M_{\text{TMM-in}}$	M_{phys}	$M_{\text{TL+phys}}$
50	8.28 ± 0.50	7.39 ± 0.67	7.21 ± 0.67	4.31 ± 0.34
100	5.60 ± 0.39	5.60 ± 0.18	5.29 ± 0.40	3.20 ± 0.24
200	3.54 ± 0.17	3.79 ± 0.42	3.59 ± 0.10	2.43 ± 0.11
350	2.45 ± 0.22	2.46 ± 0.12	2.64 ± 0.06	1.89 ± 0.02

$M_{\text{TMM-in}}$ provides only marginal improvement—+10.7% over M_0 at $n=50$, then neutral-to-negative (−0.1%, −7.0%, −0.6% at $n=100, 200, 350$)—whereas weight-level transfer with physics features ($M_{\text{TL+phys}}$) is the best model at every size, confirming that distributed weight-level transfer extracts more from the same TMM physics than direct input concatenation.

S7 Multi-fidelity Baseline Comparison

We compare PBTL against four multi-fidelity baselines: (1) linear correction, (2) residual NN, (3) Co-Kriging (Kennedy & O’Hagan autoregressive model), and (4) Co-Kriging with physics features. All baselines use 500 TMM samples and 50–350 RCWA fine-tuning samples.

Table S8: Multi-fidelity baseline comparison (Structure A, 5 seeds, 4 training sizes).

n	M_0	Linear	ResNN	Co-Krig	Co-Krig+phys	$M_{\text{TL+phys}}$ (ours)
50	8.25 ± 0.51	7.55 ± 0.08	7.83 ± 0.69	7.41 ± 0.35	7.08 ± 0.32	4.36 ± 0.29
100	5.70 ± 0.30	7.49 ± 0.10	4.77 ± 0.21	6.67 ± 0.14	6.08 ± 0.28	3.07 ± 0.24
200	3.89 ± 0.38	7.47 ± 0.04	3.46 ± 0.04	5.89 ± 0.17	4.82 ± 0.30	2.21 ± 0.05
350	2.58 ± 0.04	7.42 ± 0.01	2.45 ± 0.15	5.13 ± 0.05	3.93 ± 0.12	1.77 ± 0.03

Sparse Co-Kriging with Expanded TMM Budget

A natural concern is whether Co-Kriging would catch up if given PBTL’s 5,000-sample TMM budget. Cubic Gaussian-process (GP) scaling makes a dense fit on 2,000+ samples impractical at this budget in our implementation. We therefore replaced each per-wavelength GP with a sparse variational GP (SVGP, $M = 128$ inducing points, k -means initialization).

Table S9: Sparse SVGP Co-Kriging with expanded TMM budget (Structure A, $M = 128$ inducing points, 3 seeds).

n_{RCWA}	Sparse $N_{\text{TMM}} = 500$	Sparse $N_{\text{TMM}} = 2,000$	Dense $N_{\text{TMM}} = 500$ (ref.)	$M_{\text{TL+phys}}$
100	10.11 ± 0.52	9.96 ± 0.59	11.33 ± 0.96	7.58 ± 0.33
350	9.24 ± 0.42	9.04 ± 0.65	7.69 ± 0.31	4.85 ± 0.13

Quadrupling the TMM budget closes only 0.15–0.20 pp of the gap to PBTL, ruling out a data-budget asymmetry.

Deep Composite Multi-Fidelity NN

We additionally implemented the deep composite architecture of Meng *et al.*: a low-fidelity (LF) ResNet pretrained on 5,000 TMM samples is frozen, and a high-fidelity (HF) head predicts $y_H = \alpha(\lambda)y_L + \beta(\lambda) + F_{\text{NL}}(x, \lambda, y_L)$. The LF information enters at the output level rather than the weight level. The deep multi-fidelity neural network (MF-NN) achieves 7.18/5.24/3.48/2.59% MAE at $n = 50/100/200/350$ (3 seeds), leaving a 32–41% relative gap to $M_{\text{TL+phys}}$.

Co-Kriging Kernel Sensitivity

To rule out kernel-selection bias, we swept four kernel families and two input configurations (3 seeds): Matérn-5/2 tight, Matérn-5/2 wide, Matérn-3/2, RBF, and Matérn-5/2+phys. The best variant (Matérn-5/2+phys) reaches 12.52/10.14/8.65/7.68% MAE—a 0.1–0.6 pp improvement over the tight Matérn-5/2 but still 37–58% worse than $M_{\text{TL+phys}}$. RBF performs substantially worse (11.7–14.3%) because infinite smoothness is inappropriate for non-smooth cavity resonance features. (This kernel-sensitivity sweep was run on the earlier visible-band Structure-A pipeline and was not regenerated on the corrected pipeline; it is reported only as a kernel-family ranking, in which every Co-Kriging kernel remained well behind $M_{\text{TL+phys}}$.)

S8 Architecture Sensitivity: Per-Wavelength vs. Full-Spectrum

We re-ran M_0 and M_{TL} with a Full-Spectrum (FS) ResNet whose output is the entire 100-wavelength absorptance vector at once: input is $\text{norm}(\mathbf{x})$ (no wavelength index), the backbone is identical, and the head is replaced by a single linear layer $\mathbb{R}^{256} \rightarrow \mathbb{R}^{100}$. Pre-training uses the same 5,000-sample TMM library, and fine-tuning uses the same fixed RCWA split; the Full-Spectrum ablation is run over 3 seeds.

Table S10: Architecture sensitivity: per-wavelength vs. Full-Spectrum (FS) ResNet (Structure A). The per-wavelength columns reproduce the main-text 10-seed Structure-A result; the Full-Spectrum columns are a 3-seed ablation trained on the same fixed split.

n	Full-Spectrum (FS)		Per-wavelength	
	M_0^{FS}	$M_{\text{TL}}^{\text{FS}}$	M_0	M_{TL}
50	9.46 ± 1.12	4.92 ± 0.32	8.36 ± 0.63	4.21 ± 0.49
100	6.29 ± 0.07	3.87 ± 0.06	5.68 ± 0.33	3.11 ± 0.17
200	4.54 ± 0.16	3.04 ± 0.12	3.93 ± 0.35	2.39 ± 0.11
350	3.33 ± 0.12	2.51 ± 0.02	2.64 ± 0.09	1.82 ± 0.07

The per-wavelength architecture outperforms FS at every training size: across both M_0 and M_{TL} and all four training sizes, the FS MAE is 11–38% higher than the per-wavelength MAE (per-cell relative gap $(\text{MAE}_{\text{FS}} - \text{MAE}_{\text{PW}})/\text{MAE}_{\text{PW}}$, with the per-wavelength baseline taken from the main-text 10-seed result). PBTL pre-training nonetheless still delivers a substantial relative TL benefit on FS (+24.6% to +48.0%). Thus the weight-transfer effect is not unique to the per-wavelength output formulation, although the backbone ablation in Supplementary Section S4 shows that the benefit is not fully architecture-agnostic.

S9 Material Generalization: Au + Structure B

To test whether the Cr-vs-Au comparison generalizes beyond Structure A, we additionally ran the 4-way protocol on Structure B using corrected Johnson–Christy Au optical constants (500 targeted Au RCWA samples, 499 finite and 419 reliable after the documented physicality filter ($A \in [-0.005, 1.005]$ at every wavelength, after which A and R are clipped to $[0, 1]$), 2,000

Au TMM pre-training samples, 5 seeds per condition; the larger Au-B discard than Au-A reflects gold’s stronger oblique-incidence grazing-order sensitivity, Supplementary Section S16). After reserving 40 test and 40 validation samples, the training pool contains 339 Au RCWA samples. The Au Structure-B experiment uses gold’s native visible band (380–780 nm), where its plasmonic resonance lies, so it probes *material* rather than band generalization.

Table S11: Corrected 4-way model comparison on Au/SiO₂ Structure B (mean $\pm$ std, 5 seeds) on gold’s visible band (380–780 nm). *TL benefit* is the marginal effect of weight-level transfer alone, $(1 - \text{MAE}_{M_{\text{TL}}}/\text{MAE}_{M_0}) \times 100\%$ (M_{TL} vs. M_0 , both *without* physics features). On the corrected data the benefit is positive at every training size.

n	$M_0^{\text{Au,B}}$	$M_{\text{phys}}^{\text{Au,B}}$	$M_{\text{TL}}^{\text{Au,B}}$	$M_{\text{TL+phys}}^{\text{Au,B}}$	TL benefit (M_{TL} vs. M_0)
50	10.48 ± 0.68	10.17 ± 0.88	7.54 ± 0.30	7.57 ± 0.33	+28.0%
100	8.24 ± 0.31	7.87 ± 0.34	6.90 ± 0.42	6.93 ± 0.27	+16.4%
200	6.97 ± 0.32	6.81 ± 0.23	6.45 ± 0.21	6.33 ± 0.24	+7.5%
339	6.06 ± 0.11	6.20 ± 0.23	5.78 ± 0.08	5.73 ± 0.10	+4.6%

On the corrected Au–B data, weight-level transfer (M_{TL} vs. M_0) is positive at every training size—+28.0% at $n = 50$, +16.4% at $n = 100$, +7.5% at $n = 200$, and +4.6% at the saturated pool ($n = 339$)—weaker than Au Structure A but, unlike the earlier pre-correction visible-band check (which had turned slightly negative at the largest pool), no longer negative. Unlike Structure A, the explicit physics features give only a small standalone gain that reverses at the largest pool (M_{phys} within ~ 0.4 pp of M_0 —the planar EMA poorly captures the ring–disk Fano coupling), and the combined model satisfies $M_{\text{TL+phys}} \approx M_{\text{TL}}$; the benefit therefore comes almost entirely from weight-level transfer. The corrected TMM–RCWA fidelity is median $r = 0.78$, median MAE = 15.4%. We report this as a single descriptive cross-material result and do not draw a cross-structure mechanistic conclusion from it. Both Au checks (Structures A and B) use the deployment learning rates (scratch lr = 10^{-3} , transfer lr = 3×10^{-4}); the matched-learning-rate control that rules out a learning-rate artifact (Section S3) was performed for the three primary Cr structures and is not repeated for these secondary cross-material checks.

S10 Per-Seed Results for Structure A at n=350, 10 seeds

Table S12: Per-seed test MAE (%) for Structure A at $n = 350$ (10 seeds).

Seed	M_0	M_{phys}	M_{TL}	$M_{\text{TL+phys}}$
42	2.63	2.53	1.77	1.72
123	2.52	2.64	1.79	1.76
777	2.54	2.48	1.88	1.74
321	2.62	2.44	1.73	1.81
456	2.58	2.63	1.78	1.81
654	2.59	2.57	1.92	1.87
999	2.69	2.42	1.85	1.80
111	2.66	2.53	1.73	1.81
222	2.76	2.52	1.91	1.82
333	2.84	2.43	1.80	1.80
Mean	2.64	2.52	1.82	1.79
Std	0.09	0.07	0.07	0.04

S11 Structure C: Anisotropic TMM Pre-training Details

Structure C is inherently dual-polarization, requiring separate TE and TM absorptance predictions. We use an anisotropic TMM variant (`tmm_struct_c_aniso`) that computes separate TE and TM absorptance spectra, with $\varepsilon_{\text{eff}}^{\text{TE}}$ built from $f_y = W_y/P$ and $\varepsilon_{\text{eff}}^{\text{TM}}$ from $f_x = W_x/P$ (the anisotropic-EMA equation of the main text). Five additional polarization-specific features are included: (1) the directional fill fraction $f_x = W_x/P$, (2) the directional fill fraction $f_y = W_y/P$, (3) the normalized geometric anisotropy $|f_x - f_y|/(f_x + f_y)$, and the polarization-resolved sub-wavelength resonance ratios (4) $W_x^2/(P\lambda)$ and (5) $W_y^2/(P\lambda)$. The total physics feature count is 18 (13 base + 5 polarization).

The dual-polarization architecture places two independent sigmoid heads on the shared 256-dimensional representation, predicting R_{TE} and R_{TM} ; absorptances are derived deterministically as

$$A_{\text{TE}} = 1 - R_{\text{TE}}, \quad A_{\text{TM}} = 1 - R_{\text{TM}},$$

and transmittance is negligible (Section S12) and not predicted. The training loss is the sum of the four per-channel MSE terms on $(A_{\text{TE}}, R_{\text{TE}}, A_{\text{TM}}, R_{\text{TM}})$.

S12 Transmittance Validation

Table S13: Transmittance statistics from RCWA simulations across all three MIM structures (reliable-sample subset), confirming that the Cr ground mirror keeps transmittance small: per-structure $T_{\text{mean}} < 1\%$ and $T_{\text{max}} \lesssim 2.3\%$, so the structures operate as near-opaque absorbers. The RCWA training labels are the full absorptance $A = 1 - R - T$; because the surrogate predicts R and outputs $A = 1 - R$, this residual transmittance bounds the absorptance-output approximation error—at most T per sample (mean ~ 0.1 – 0.6% , up to $\sim 2.3\%$ worst-case), as quantified in the main-text Neural Network Architecture section—small relative to the reported MAE at small and medium training pools though not negligible against the smallest MAE values at the largest pools.

MIM Structure	T_{mean}	T_{max}	% with $T < 0.01$
A (Dual-Cavity)	0.0013	0.0142	99.5%
B (Ring-Disk)	0.0061	0.0209	71.6%
C-TE (Dual-Pol)	0.0046	0.0224	83.0%
C-TM (Dual-Pol)	0.0054	0.0234	78.5%

S13 Preliminary Inverse Design Investigation

We attempted inverse design using the trained PBTL surrogate ($M_{\text{TL+phys}}$, Structure A) with gradient-based optimization. This probe was run on the earlier visible-band (380–780 nm, fixed $N=5$) surrogate and is reported as a qualitative, preliminary demonstration; the adversarial-gradient failure mode it illustrates is reported as an illustrative example on that superseded visible-band surrogate and was not revalidated on the corrected pipeline. Both RCWA solvers showed significant validation failures:

These failures reflect extrapolation limits of the surrogate model and fundamental incompatibilities between RCWA solver implementations. The cross-solver discrepancy is confined to the inverse design context; it does not affect the forward modeling results, which use only `torcwa` for both training and evaluation within the design space. Future work should employ physics-informed inverse methods with constrained optimization to prevent adversarial exploitation of surrogate gradients.

Table S14: Inverse design validation failure analysis.

Solver	Result	MAE	Root Cause
torcwa	Complete failure	$A \approx 0$	Adversarial exploitation of surrogate gradient
grcwa	High error	24–41%	Solver parameter mismatch + extrapolation
grcwa (sanity)	Failed reproduction	12.22%	Fundamental solver incompatibility

S14 Sub-band TMM–RCWA Fidelity Decomposition for Structures B and C

To support the wavelength-region-dependent fidelity interpretation in the cross-structure analysis of the main text and the Structure-B characterization there, we decompose the corrected TMM–RCWA fidelity of Structures B and C into spectral sub-bands. The same 500-sample RCWA/TMM pairs used for the main-text TMM–RCWA fidelity table are partitioned by wavelength, and per-sample Pearson correlation r and absolute MAE are recomputed restricted to each sub-band; the structures’ full operating band (400–1800 nm) is also reported for reference.

Table S15: Sub-band TMM–RCWA fidelity for Structures B and C (each on the broadband 400–1800 nm native grid; up to 500 samples per structure (500/499/500 for A/B/C)). Median Pearson r and median absolute MAE are reported per sub-band. The deep-NIR rows isolate the sub-wavelength regime where the EMA approximation underlying TMM is expected to be most accurate, while the visible/short-wavelength rows isolate the region where Fano coupling (Structure B) and polarization-resonance splitting (Structure C) are strongest. The Full-broadband row reports the per-sample *median* MAE and matches the median values reported in the main-text TMM–RCWA fidelity table.

Sub-band	B median r	B median MAE	C median r	C median MAE
Full broadband (400–1800 nm)	+0.96	8.9%	+0.65	16.9%
Visible (400–780 nm)	+0.95	8.2%	+0.62	12.5%
NIR-short (780–1200 nm)	+0.97	9.7%	+0.64	17.3%
NIR-long (1200–1800 nm)	+ 1.00	8.9%	+0.71	18.4%
Deep NIR (1000–1800 nm)	+0.99	9.0%	+0.63	18.2%

Interpretation. On the corrected materials and grid, Structure B shows *uniformly high* TMM–RCWA agreement: the median r is 0.95–1.00 in every sub-band and the median MAE stays within ~ 8 –10% throughout. The earlier, apparent visible-band “shape degradation” was an artifact of the uncorrected optical constants; with the corrected Johnson & Christy data the EMA-based TMM tracks the shape of Structure B’s spectra across the whole operating band, with an operating-band MAE of ~ 8 –10%, consistent with Structure B showing a positive weight-transfer benefit (main-text Structure-B/C results). Structure C is moderate across all sub-bands (median $r \approx 0.62$ –0.71, MAE ≈ 12 –18%); its lower overall fidelity—driven by the incomplete polarization splitting of a planar EMA source—is consistent with its being the smallest-benefit structure.

These sub-band measurements support the revised main-text narrative: the cross-structure benefit ordering follows the absolute operating-band MAE ($A < B < C$) rather than shape correlation r alone, and Structure B in particular is now a *high-fidelity* source in every sub-band rather than a visible-band failure case (main-text cross-structure analysis and guidelines).

Common operating band. In the submitted version Structure A was simulated on a visible 380–780 nm grid while Structures B and C used 400–1800 nm, raising a visible-versus-broadband fairness question. In the revised pipeline Structure A is unified to the same 400–1800 nm band as B and C (Methods), so all three structures are now compared on a common

broadband grid and that asymmetry no longer applies; the visible (400–780 nm) sub-band cut above (B +0.95/8.2%, C +0.62/12.5%) is reported for completeness, supporting the joint r -and-MAE diagnostic recommended in the main-text practical-guidelines discussion.

S15 Head-to-Head with Peng et al.’s PCA-Centroid Indicator

Reviewer point M8 asks for a direct comparison with the PCA-projected far-field centroid distance proposed by Peng et al. (main-text reference). We implemented their indicator on our own data: for each condition we fit PCA on the pooled (TMM, RCWA) absorptance spectra and take the Euclidean distance between the TMM and RCWA cluster centroids in the principal-component space (95% variance retained), and we benchmark it against our joint pilot-set r -and-MAE diagnostic as a predictor of the measured weight-level transfer benefit. The benchmark spans nine conditions: the three Cr structures and the six controlled noise-injection levels on Structure A (the noise-injection experiment of the main text).

Table S16: Per-condition values for the head-to-head. “Peng dist.” is the PCA-centroid distance (95% variance); “our r ” is the median per-sample TMM–RCWA shape correlation (the same statistic used in the main-text noise table); “benefit” is the $M_{\text{TL-v}s-M_0}$ weight-transfer benefit at $n = 50$.

Condition	Peng dist.	our r	our MAE (%)	benefit (%)
A (Structure)	0.74	+0.83	7.9	+49.7
B (Structure)	0.38	+0.96	8.9	+36.3
C (Structure)	1.62	+0.65	16.9	+35.8
noise $\sigma = 0.00$	0.85	+0.77	7.4	+47.3
noise $\sigma = 0.05$	0.86	+0.50	8.0	+40.8
noise $\sigma = 0.10$	0.88	+0.35	10.8	+33.0
noise $\sigma = 0.15$	0.92	+0.26	14.0	+32.3
noise $\sigma = 0.20$	0.98	+0.21	16.7	+22.6
random ($\sigma = \infty$)	2.52	−0.03	33.0	−57.9

Table S17: Predictive power (Pearson correlation with benefit) of each indicator, on the three structures, the six noise levels, and all nine conditions. Lower (more negative) is a stronger predictor for the distance/MAE; positive for r . The shape-correlation r here is the median per-sample statistic (consistent with the main-text noise table), not the raw spectrum-level r .

Predictor	3 structures	6 noise	all 9
Peng centroid distance	−0.26	−0.99	−0.86
our MAE (amplitude)	−0.61	−0.98	−0.95
our r (shape)	+0.12	+0.81	+0.73

The operating-band *absolute MAE* is the strongest single predictor of the weight-transfer benefit across all nine conditions ($|r| = 0.95$; rank correlation -1.0 across the three structures), ahead of both the Peng centroid distance ($|r| = 0.86$ for the published raw, 95%-variance definition; the strongest of the four Peng PCA variants we evaluated reaches $|r| \approx 0.93$, still below the MAE’s 0.95) and the median per-sample shape correlation r alone ($|r| = 0.73$ combined). The amplitude–shape separation is sharpest on the controlled noise sweep: there the MAE attains $|r| = 0.98$ while the median shape- r reaches only $|r| = 0.81$, even though the underlying sweep is monotone in fidelity (all three predictors hold rank correlation $|\rho| = 1$). The rank agreement confirms that each indicator is a correctly-signed monotone marker, but the linear gap shows that the benefit scales with the amplitude error far more tightly than with the shape correlation. The discriminating regime is the three real structures, where the MAE ordering matches the

benefit ordering exactly (rank correlation -1.0) while both the Peng distance and the shape- r under-rank them ($B > A > C$ rather than $A > B > C$). The three-structure *linear* (Pearson) correlation is modest for every predictor at this small training size—at $n = 50$ all three structures, including the lower-fidelity Structure C, receive large transfer gains, compressing the amplitude-benefit spread—so we read the three-structure result through its rank ordering (where MAE alone is monotone, $\rho = -1.0$) rather than the three-point Pearson coefficient ($|r| = 0.61$ for MAE, $0.26/0.12$ for the Peng distance/shape- r); the linear amplitude-benefit gap is established instead by the controlled six-level noise sweep ($|r| = 0.98$ vs. 0.81) and the all-nine aggregate ($|r| = 0.95$). We emphasize that the Peng distance is a valid, correctly-signed monotone indicator; the difference is that our amplitude (MAE) component is the stronger ordering signal in this cross-fidelity, amplitude-dominated regime, and our diagnostic additionally requires neither a PCA projection nor a pre-trained model. The comparison is illustrative at $N = 3$ structures and descriptive throughout (no significance is claimed).

S16 RCWA Fourier-Truncation Convergence

Reviewer point M7 notes that the RCWA forward simulations were run at a fixed, modest Fourier-truncation order and that the resulting truncation error was absorbed into the reported surrogate MAE. To quantify this and to justify the settings used for the regenerated dataset, we performed a per-(geometry, wavelength) convergence study, sweeping the truncation order $N \in \{5, 9, 13, 17\}$ (the number of retained Fourier harmonics per in-plane direction) and tracking the absolute absorptance. For the periodic two-dimensional Structure-C geometries we resolve the two linear polarizations separately (TE and TM); for Structures A and B we report the unpolarized absorptance.

The headline finding is that $N = 5$ —the order used in the original submission—is under-converged, and the residual truncation error is structure-dependent. Comparing $N = 5$ against the converged $N = 17$ reference, the absolute-absorptance difference is ≈ 2.1 pp for Structure A (up to ~ 5.7 pp at long wavelengths), ≈ 3.4 pp for Structure B, and ≈ 5.5 pp for Structure C on the TE channel, growing to as much as ~ 15 pp at long wavelengths ($\lambda \approx 1000$ nm), where the high-aspect-ratio anisotropic grating profile of Structure C requires the most harmonics to resolve. The TM channel of Structure C is comparatively benign ($\lesssim 1$ pp at the canonical geometry). Representative (structure, λ) pairs are given in Table S18.

This accuracy comes at a steep cost. The per-(geometry, wavelength) wall-clock time grows from ≈ 0.4 s at $N = 5$ to ≈ 28 s at $N = 17$ (roughly a $70\times$ increase), and the peak memory grows from ≈ 77 MB to ≈ 7 GB over the same range. The dataset regeneration therefore uses *per-wavelength adaptive* Fourier orders, up to $N = 17$, raised by a heuristic based on feature size and anisotropy rather than held at a single fixed order: short wavelengths and the low-anisotropy control geometries are assigned lower N , while the long-wavelength, high-aspect-ratio Structure-C geometries are pushed to the maximum order ($N = 17$, the practical memory ceiling), at which a small residual truncation can persist. As a result, the surrogate MAE reported in the main text is dominated by the network’s approximation error rather than RCWA truncation error—with any residual applied identically across the model variants compared—which directly addresses reviewer point M7.

Note. The $N = 17$ reference for the long-wavelength rows is the highest order reached before the memory ceiling; where $N = 17$ was infeasible the deepest converged order ($N = 13$) is used as the reference, so the tabulated Δ is a conservative (lower-bound) estimate of the $N = 5$ truncation error.

Table S18: Representative RCWA Fourier-truncation convergence. “ $A(N=5)$ ” and “ $A(N=17)$ ” are the absolute absorptances at the under-converged and converged orders; Δ is their difference in percentage points. Structure-C entries are on the TE channel, where truncation error is largest; the long- λ rows show the worst-case residual. The Structure-A 380 nm entry is a deliberate below-band short-wavelength stress point (Structure A’s operating band is 400–1800 nm); the in-band Structure-A behavior is given by the 1000 nm row.

Structure	λ (nm)	$A(N=5)$	$A(N=17)$	Δ (pp)
A (wide-aspect)	380	0.406	0.385	2.1
A (wide-aspect)	1000	0.943	0.886	5.7
B (ring-disk)	400	0.525	0.491	3.4
C (TE, canonical)	400	0.466	0.412	5.5
C (TE, canonical)	1000	0.761	0.608	15.3

Oblique-Incidence Power Normalization

A second, distinct source of unphysical spectra at oblique incidence is the diffraction-order power normalization rather than Fourier truncation. At high incidence angle, grazing propagating orders (diffraction angle approaching 90°) are ill-conditioned in the `torcwa` longitudinal-field power normalization, which can inflate their apparent diffraction efficiency and yield $R + T > 1$ (equivalently $A = 1 - R - T < 0$) for a minority of high-angle geometries. We exclude near-grazing orders—those whose propagation cosine $\text{Re}(k_z)/n$ falls below a cutoff $\text{COS}_{\min}$ —from the power sum; the production datasets use $\text{COS}_{\min} = 0.10$ (diffraction angle $> 84^\circ$), and the small residual that survives this cutoff is removed by the per-structure reliability filter. On the affected samples, tightening the cutoff to $\text{COS}_{\min} = 0.30$ (excluding orders beyond $\sim 72^\circ$) eliminates the residual $R + T > 1$: for example, the worst Structure-B oblique sample drops from $R + T = 1.81$ to 0.83, while high-angle samples that were already physical are left essentially unchanged (e.g. a $\theta \approx 60^\circ$ good sample keeps the same absorptance for every cutoff tested). This confirms that the discarded spectra are a grazing-order normalization artifact, not a physical or convergence effect. The reported datasets retain the $\text{COS}_{\min} = 0.10$ filter, so these artifact samples are simply excluded and do not enter any reported metric.

Footprint of the cutoff on the reported data. To bound how much the cutoff choice could affect the reported metrics, we analytically audited every reliable (sample, wavelength) entry, enumerating the propagating diffraction orders from the grating equation—both reflection orders in air and transmission orders in the glass substrate ($n=1.5$)—and flagging those in the cutoff-sensitive near-grazing band. The *reflection-side* orders, which drive the $R + T > 1$ artifact, fall in the cut range for only 0.35%, 0.95%, and 0.99% of reliable entries (Structures A, B, and C) at the production cutoff $\text{COS}_{\min} = 0.10$. Including *transmission-side* grazing orders raises the union to 1.5%, 3.6%, and 3.7% at the production cutoff (and to 11%, 22%, 22% if the cutoff were widened to 0.30); however, these transmitted orders can carry at most the total transmittance, which is $\leq 2.4\%$ ($T_{\max} \approx 2.3\%$) for these mirror-backed stacks, so their possible contribution to the absorptance is bounded by this negligible transmittance. Every affected order lies beyond a 72° diffraction angle—the regime where the `torcwa` normalization is unreliable. Because the identical cutoff and reliability filter are applied to all four model variants within each structure, the M_{TL} -vs- M_0 comparisons that underlie our conclusions are independent of the cutoff choice. (For the released code we floor $\text{COS}_{\min}$ above zero, since $\text{COS}_{\min} = 0$ would retain exactly-grazing orders whose vanishing longitudinal wavevector destabilizes the power normalization.)

S17 TMM Pre-training Geometry-Constraint Sensitivity

The TMM pre-training libraries are generated by uniform sampling over the design-parameter ranges, whereas the RCWA fine-tuning and test geometries additionally satisfy the structure-specific fabrication constraints ($W_x, W_y, W_z \leq 0.9P$ for Structures A and C; $R_{\text{out}} \leq 0.45P$ together with $R_{\text{in}} < R_{\text{out}}$ and $R_{\text{disk}} < R_{\text{in}}$ for the ring-disk Structure B). A fraction of the pre-training samples therefore lie outside the constrained design region. To verify that this does not inflate the reported weight-transfer benefit, we re-ran the M_{TL} -vs- M_0 comparison with the TMM pre-training samples projected onto the same constraints, holding all other settings (architecture, training schedule, seeds, and fine-tuning data) fixed.

Across all three structures and both low-data regimes the constrained and unconstrained pre-training libraries produced similar benefits (Table S19): the mean weight-transfer benefit changed by less than 2 percentage points for Structures A and B at both training sizes and for Structure C at $n = 50$. The single larger change, Structure C at $n = 350$ ($+4.3\% \rightarrow +9.8\%$), is in the direction of a *larger* benefit under the constrained library—unconstrained sampling, if anything, slightly *under*-states rather than inflates the gain. We therefore find no evidence that the reported PBTL benefit is an artifact of unconstrained TMM sampling.

Table S19: Weight-transfer benefit $1 - \text{MAE}(M_{\text{TL}})/\text{MAE}(M_0)$ (mean over three seeds) with unconstrained vs. constrained TMM pre-training sampling. Differences are within seed-to-seed noise except for Structure C at $n = 350$, where the constrained library gives a *larger* benefit.

Structure	n	Unconstrained	Constrained	Δ
A	50	+54.9%	+53.3%	−1.6 pp
A	350	+30.7%	+32.5%	+1.8 pp
B	50	+37.4%	+36.8%	−0.5 pp
B	350	+9.7%	+8.0%	−1.7 pp
C	50	+24.4%	+25.8%	+1.4 pp
C	350	+4.3%	+9.8%	+5.5 pp

S18 Additional Figures (Reviewer #2 Requests)

The following figures address specific requests from Reviewer #2 (Figures S1–S4).

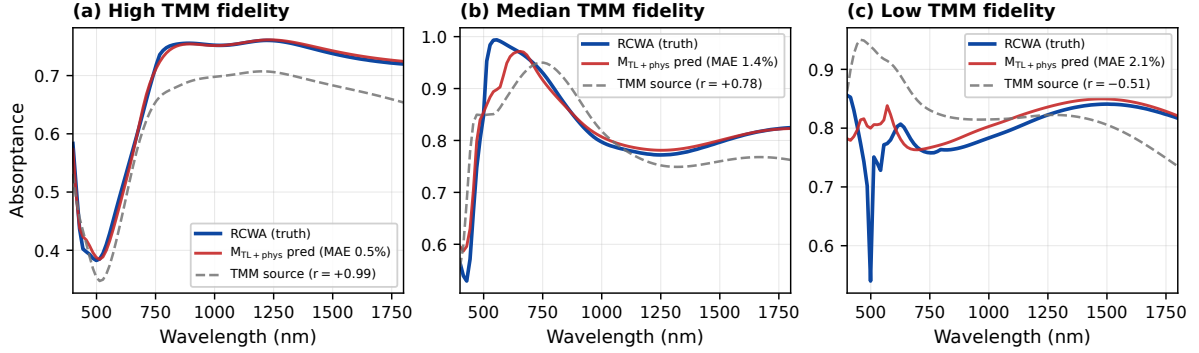


Figure S1: Fine-tuned surrogate ($M_{\text{TL}+\text{phys}}$, Structure A, $n = 350$) predicted absorbance vs. RCWA ground truth for representative test geometries spanning high, median, and low TMM–RCWA fidelity (per-sample shape correlation r indicated). The fine-tuned surrogate tracks the full-wave RCWA closely (per-sample MAE 0.5–2.1% for these representative geometries) even where the TMM pre-training source (dashed) diverges from RCWA, confirming that fine-tuning on RCWA labels corrects the source’s amplitude error.

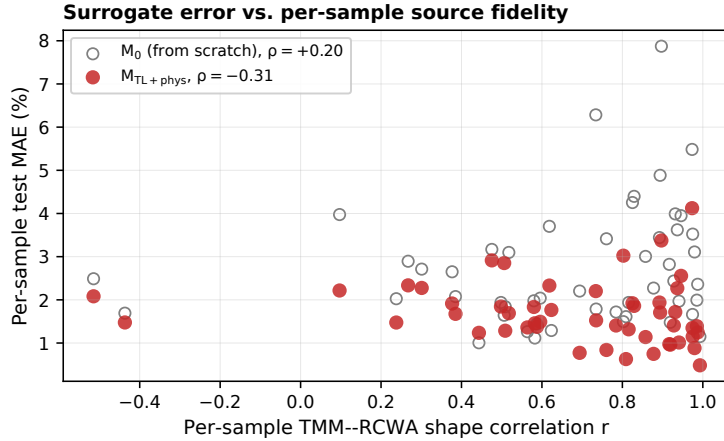


Figure S2: Per-sample surrogate test MAE vs. per-sample TMM–RCWA shape correlation r (Structure A, 50 test geometries at $n = 350$). The fine-tuned $M_{\text{TL}+\text{phys}}$ error is only weakly related to the per-sample source fidelity (Spearman $\rho = -0.31$) and is below the from-scratch M_0 for 44 of the 50 test geometries (median 1.5% vs. 2.3%): the source-fidelity dependence of PBTl lives in the *aggregate* weight-transfer benefit, not in a per-test-point error penalty.

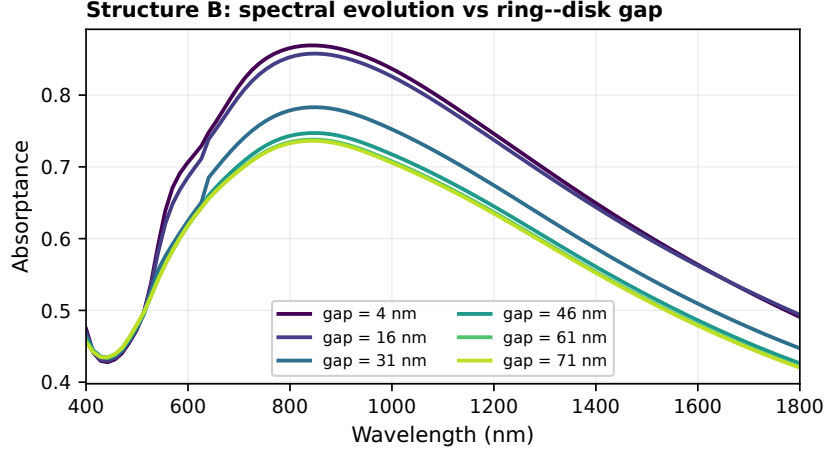


Figure S3: Structure B (ring-disk) RCWA absorptance as the ring-disk gap is varied (disk radius swept at fixed ring geometry). The absorption resonance weakens monotonically as the gap widens (peak absorbance $0.87 \rightarrow 0.74$), supporting the interpretation that the Structure-B response arises from ring-disk near-field coupling. The sweep varies the disk radius, so it probes the coupled response rather than the gap in isolation.

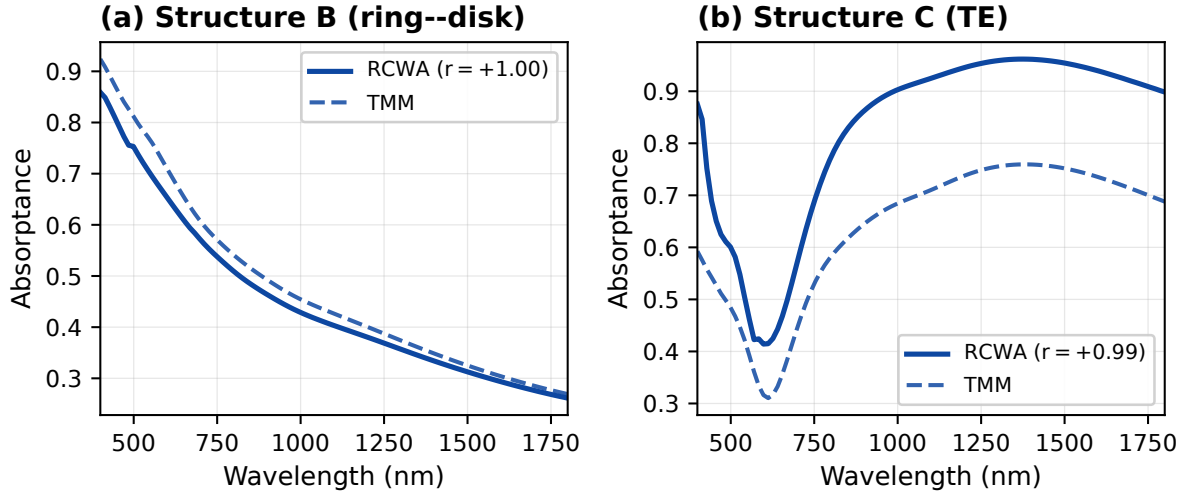


Figure S4: Representative high-fidelity TMM (dashed) vs. RCWA (solid) absorptance spectra for Structures B and C (Structure B by highest per-sample r ; Structure C the TE-polarized channel, selected by highest TE-only r), illustrating the cavity-like responses that the planar TMM source captures where its fidelity is high.

S19 Correction Change-Log

While preparing the response to Reviewer #1, the convergence comment (point M7) prompted a full audit of the forward-simulation pipeline. The audit identified four corrections; because two of them required rebuilding the simulation data, we regenerated the complete RCWA dataset for all three structures (A, B, C) and re-ran every downstream analysis. Table S20 catalogues the corrections, and Table S21 reports the corrected Table I (TMM–RCWA spectral fidelity) against the submitted values. The optical-constant correction is the dominant driver of the change.

Table S20: The four corrections applied during the pre-decision audit.

Correction	Description
(1) Fourier convergence (requested, M7)	Under-converged truncation ($N = 5$) replaced by per-wavelength adaptive orders up to $N = 17$ (raised by a feature-size/anisotropy heuristic; $N = 17$ is the practical memory ceiling, Sec. S16); the reported surrogate MAE is dominated by network error rather than RCWA truncation, with any residual applied identically across model variants.
(2) Optical constants (independent)	Tabulated Cr/Ti/Au extinction coefficients were inconsistent with the cited Johnson & Christy source (off by $\sim 40\text{--}70\%$, up to ~ 36 pp absorption impact in some regimes); rebuilt on measured Johnson & Christy constants (metals) and Siefke constants (TiO_2); both source datasets are cited in the main-text Methods (“Optical constants”). <i>Dominant driver of the Table I change.</i>
(3) Wavelength-grid fix (independent)	The Table I fidelity script compared TMM and RCWA spectra on mismatched wavelength grids; corrected to each structure’s native grid, with Structure A unified to the broadband 400–1800 nm convention used for B and C.
(4) Structure-C bounds (independent)	The W_x/W_y normalization bounds were $[50, 400]$ nm, narrower than the declared design space ($[50, 720]$ nm); 52% of Structure-C samples normalized outside $[0, 1]$ and the TMM pre-training library under-covered the design space. Bounds corrected to the declared range and all Structure-C results re-run; the weight-transfer benefit grows from +4% to +9.7% at $n = 350$.

Table S21: Corrected Table I (TMM–RCWA spectral fidelity), submitted vs. corrected. “Median r ” is the median per-sample shape correlation; “TMM MAE” is the operating-band absolute error. Values reflect all corrections jointly; the optical-constant fix dominates.

Structure	Median r (subm.)	Median r (corr.)	TMM MAE (subm.)	TMM MAE (corr.)
A	+0.72	+0.83	15.0%	7.9%
B	−0.07	+0.96	27.0%	8.9%
C	+0.34	+0.65	21.3%	16.9%

Impact on the conclusions. These corrections do more than rescale magnitudes, and we state the effect plainly. On the corrected data, the central contribution—the fidelity–benefit diagnostic—continues to be supported, though the supporting evidence changed: across the three corrected structures the weight-transfer benefit forms a monotone gradient ($A > B > C$) that is ordered by the operating-band absolute MAE rather than by shape correlation r (the median- r ordering $B > A > C$ mis-ranks the benefit). We note this is a cross-structure ordering over only $N=3$ points (cross-structure linear correlation $|r| = 0.61$), complemented by a

single-structure controlled noise sweep; we do not claim the original conclusions carried through the corrections unchanged. One qualitative conclusion has changed: on the corrected data Structures B and C no longer show natural negative weight-level transfer—both are positive at every training-set size—so the original “two distinct negative-transfer failure modes” taxonomy has no empirical support and has been *removed* from the manuscript. Negative transfer is retained solely as a *controlled* result: degrading the TMM source toward zero fidelity drives the benefit from +47% at full fidelity to −58% at zero fidelity, a monotone fidelity–benefit relationship (Spearman $\rho = -1.0$) in which the operating-band MAE predicts the benefit far better than the shape correlation r ($|r| = 0.98$ vs. 0.81 on the noise sweep; Sec. S15).